Worksheet # 6 Ch 6.1

6.1 Normal Probability Distribution

6.1 For a normally distributed population with $\mu = 200$ and $\sigma = 20$, determine the standardized z-value for each of the following.

a. $x = 225$

b. $x = 190$

c. $x = 240$

6.2 For a standardized normal distribution, calculate the following probabilities:

a. $p(z < 1.5)$

b. $p(z \geq 0.85)$

c. $p (-1.28 < z < 1.75)$

Worksheet # 6 Ch 6.1

6.4 For a standardized normal distribution, determine a value, say z_0, so that

a. p(0 < z < z_0) = 0.4772

b. p (-z_0 ≤ z < 0) =0.45

c. p (-z_0 ≤ z ≤ z_0) = 0.95

d. p (z > z_0) = 0.025

e. p (z ≤ z_0) = 0.01

Worksheet # 6 Ch 6.1

6.7 For the following normal distributions with parameters as specified, calculate the required probabilities:

a. $\mu = 5$, $\sigma = 2$; calculate $p(0 < x < 8)$

b. $\mu = 0$, $\sigma = 3$; calculate $p(x > 1)$

6.13 A random variable is normally distributed with a mean of 60 and a standard deviation of 9.

a. What is the probability that a randomly selected value from the distribution will be less than 46.5 ?

b. What is the probability that a randomly selected value from the distribution will be greater than 78 ? (self-study)

c. What is the probability that a randomly selected value will be between 51 and 73.5 ?

Worksheet # 6 Ch 6.1

6.17 The average number of acres burned by forest and range fires in a large Wyoming county is 5300 acres per year, with a standard deviation of 750 acres. The distribution of the number of acres burned is normal.

a. compute the probability that more than 6000 acres will be burned in any year.

b. Determine the probability that fewer than 5000 acres will be burned in any year.

c. What is the probability that between 3500 and 5200 acres will be burned?

d. In those years when more than 6500 acres are burned help is needed from eastern region fire teams. Determine the probability that the help will be needed in any year.